

Apex Grand Prix Racing League

Database Project

The Scenario

The Apex Grand Prix is a global open-wheel racing championship, run across a calendar of circuits spanning deserts, coastlines, mountains, and city streets. Ten teams field four drivers between them, and every single lap of every race — practice, qualifying, and the race itself — is timed to the thousandth of a second by trackside sensors.

That timing data feeds a live database used by race engineers, team strategists, the broadcast production crew, and the league's own commercial and technical departments. Split-second lap time differences decide championships, pit strategy calls are made from this data in real time, and every team principal wants their numbers before their nearest rival does.

The database is built around five connected tables:

- Teams — every constructor competing in the championship
- Drivers — every driver on the grid, linked to the team they race for
- Circuits — every track on the calendar
- Races — every individual race held at a circuit, across multiple seasons
- LapTimes — every single lap recorded by every driver in every race

Altogether the database holds 30,000 records across these five tables — the overwhelming majority of them individual lap times, since a single race weekend alone generates thousands of them.

Below are 75 real situations that come up in running a championship like this. Read each one and work out how you'd solve it using the database — nothing is labelled, so the first step is always deciding what the situation is actually asking you to do.

Cases

1. For each driver, looking at their laps in order, show what their lap time was on the lap immediately before the current one — this helps spot a sudden pace drop, possibly from traffic or a mistake.
2. The strategy team wants something they can plug straight into their live timing tool that always reflects each team's current total pit stop count, without re-deriving it manually each session.
3. Identify any driver who has never completed a single lap this season — a serious performance concern to flag to the team.
4. Count the laps completed by each driver this season, and shortlist only the drivers with fewer than 100 total laps recorded — they may have suffered reliability issues.
5. Going forward, the league wants to record each driver's contract expiry date. Adjust the driver records so this detail can be captured.
6. Group all lap times by circuit, and flag circuits where the average lap time is unusually high — this may point to a particularly technical or slow layout.
7. Prepare a results sheet listing every lap along with the driver's name and the team they race for, all on a single line per lap.
8. The broadcast team wants to know, in one number, how many pit stops have been logged across the entire season.
9. Build an intermediate circuit-by-circuit average lap time, then use it to flag every circuit where laps are running noticeably slower than the season-wide average.
10. For each driver, show what their very next lap time will be, placed right alongside the current one, to help spot a pattern building up across consecutive laps.
11. List every driver along with their most recent lap time, making sure drivers who haven't completed a single lap this weekend still appear on the list rather than disappearing.
12. Engineers constantly search lap records using a specific driver's ID to pull up their full season history instantly, but this lookup has started to feel sluggish now that the lap table has grown into the tens of thousands. What would you do about it?
13. A batch of very old test-session lap times from a discontinued practice format is cluttering the database and is no longer needed for any analysis. Remove exactly those records.
14. Work out how many distinct circuits the championship has raced at so far.
15. Label each lap as 'Fast' (under 85 seconds), 'Average' (85-95 seconds), or 'Slow' (above 95 seconds), so the strategy team can spot pace trends at a glance.
16. Across the entire season, order every lap by lap time to surface the fastest laps of the year, making sure that if two laps share the exact same time, neither one gets unfairly skipped in the ordering.
17. A timing system recalibration means every lap time recorded at one specific circuit needs to be corrected by a fixed offset. Apply this correction to just that circuit's laps.

- 18.** Track down which circuit hosted the race with the single fastest average lap time recorded anywhere this season.
- 19.** The scheduling team wants to know which day of the week races are most commonly held on, to help plan next year's calendar.
- 20.** Cross-check circuits against the race calendar to identify any circuit that hasn't been used in the current season at all.
- 21.** Identify all races that took place more than two years ago, since historical data beyond this point is being moved to a separate archive.
- 22.** A data-entry mistake caused an entire race's laps to be logged under the wrong circuit altogether. Correct this in one sweep rather than lap by lap.
- 23.** The marketing team wants to flag any circuit name that's unusually long (more than 25 characters) so it can be shortened for the mobile app.
- 24.** A freelance analyst's contract for the season has just ended, and their access to the database needs to be fully withdrawn with immediate effect.
- 25.** An engineer tries to save a new driver record without assigning them to a team. What should prevent that incomplete record from being saved?
- 26.** A new lap record arrives for a race that was never added to the calendar. Explain what should happen to this record and why it must not be allowed to slip through.
- 27.** A driver who was thought to be out for the season due to injury has just been confirmed to return for the next race. Update their team assignment to reflect a new team signing.
- 28.** Work this out in two stages: first tally up each driver's total number of laps this season, then use that to shortlist drivers who've completed fewer than 200 laps — these may have missed races.
- 29.** Sort each team's founding year into 'Veteran' (before 2000), 'Established' (2000-2012), or 'New Entrant' (after 2012), to help commentators frame the grid's history on air.
- 30.** Within each race, order the drivers by their lap times, fastest first, to reconstruct the effective running order for that lap.
- 31.** For the technical review, work out the average lap time recorded at each individual circuit.
- 32.** After a few seasons and some team changes, nobody quite remembers what was set up to speed up lookups on the lap times table. Check what's currently in place.
- 33.** For a driver under investigation for a scheduling dispute, work out how many days passed between their first race of the season and their most recent one.
- 34.** Calculate the average number of championships won per team, to see how competitive the grid has historically been.
- 35.** Identify every driver who belongs to whichever single team currently has the most championships won.

- 36.** The chief timing officer needs the ability to correct lap time records directly, unlike the rest of the analysis staff. Set up an access level that reflects this difference.
- 37.** Identify which drivers have posted an average finishing position better than 5 across all their races this season — genuine championship contenders.
- 38.** Flag each circuit as 'High-Speed', 'Technical', or 'Balanced' based on its total lap count relative to its length, to help engineers set up the car correctly.
- 39.** Find every team that has at least one driver who has posted a lap time at a circuit outside their home country.
- 40.** First tally each team's total pit stop count, then bring in driver details to see which drivers are contributing most to their team's total pit stops.
- 41.** For the TV broadcast graphics, combine each driver's name and car number into a single readable label, separated by a dash.
- 42.** List all races held at one of the three longest circuits by track length.
- 43.** Group all races by the exact season year they belong to, to compare how the calendar has grown year over year.
- 44.** The broadcast graphics team needs a simplified, always-ready snapshot showing each driver's name, team, car number, and latest lap time, without the production crew having to reconstruct the full picture from scratch every race weekend.
- 45.** Calculate a combined performance score per driver using their average lap time and finishing position as an intermediate result, then use that to rank the teams by overall performance.
- 46.** Based on finishing position, tag each lap record as 'Podium Pace', 'Points Pace', or 'Midfield Pace', to help analysts describe a driver's race in simple terms.
- 47.** Count how many drivers are currently contracted to each team.
- 48.** For a specific circuit hosting a night race, pull all races and check which ones started particularly late in the season.
- 49.** Some circuits may not have hosted a single race yet this season. Identify exactly which ones, since this is a scheduling gap worth flagging.
- 50.** The end-of-season report needs a single consolidated result: for every team, the number of drivers on their roster, their average lap time across the season, their total pit stop count, and a clear flag for any team whose average lap time has fallen below the grid-wide average.
- 51.** Find the single fastest lap ever recorded across the whole season — this could be the benchmark used for next year's track record board.
- 52.** Split all drivers into three roughly equal performance tiers based on their average lap time, to help design three different tiers of prize money.

- 53.** Pull every lap time belonging specifically to the one driver who recorded the single fastest lap of the season.
- 54.** Some driver names were typed with inconsistent capitalization during registration. Produce a clean, uniformly capitalized list for the official entry list.
- 55.** After a recent data cleanup, a few lap records may now point to a driver ID that no longer exists in the system. Track down these orphaned records.
- 56.** For each team, count the number of pit stops logged across the season, and highlight any team with more than 150 — a possible sign of strategy issues or reliability trouble.
- 57.** Produce a single combined list of every circuit and every race together — including circuits with zero races and any race record that's somehow missing its circuit — so nothing gets left out of the audit.
- 58.** Break this into two steps: first find each driver's most recent race date, then use that to identify which drivers haven't raced in the last two rounds.
- 59.** Group laps by race and identify which races had an average lap time slower than the circuit's typical pace — a possible sign of poor weather conditions.
- 60.** For each team, build a running count of pit stops over the course of the season, ordered by race date, to visualize how strategy has evolved.
- 61.** Find the slowest lap time ever recorded — this might point to a driver who had to nurse a damaged car home.
- 62.** The system currently lets someone log a finishing position of 47 in a 20-car race by mistake. What change would permanently make this impossible going forward?
- 63.** For each driver, flag any of their own laps where the recorded lap time is noticeably slower than that same driver's own typical pace — a possible sign of a spin or damage.
- 64.** A data entry clerk accidentally types a lap time as a negative number. How do you make sure this kind of entry is rejected the moment it's submitted?
- 65.** For every race, surface the single fastest lap recorded in it without losing any of that race's other lap records in the process — every lap should still show up in the result.
- 66.** For each circuit, smooth out lap-to-lap variation by calculating an average lap time using only the current lap and the two immediately before it.
- 67.** For every team on the grid, list out all the circuits where at least one of their drivers has posted a lap time.
- 68.** The live timing dashboard constantly filters by race and lap number together, and the two filters are almost always used side by side. How would you set the database up so this specific combined search runs faster?
- 69.** Pull every race that took place during the month of June across all seasons, to study how mid-season form typically shifts.

70. The league commissioner wants a single number for the season wrap-up: exactly how many laps have been recorded across the entire championship so far.

71. For each driver, work out how many days have passed since their date of birth — the broadcast team wants live age figures on screen during the race.

72. The commercial partnerships team keeps asking for the same list — every team along with its base country and championship count. Set up something they can pull directly, without you explaining the underlying tables every time they ask.

73. A bulk import left several circuit names with accidental leading or trailing spaces. Clean these up before they go into the printed race programme.

74. A new data analyst is joining the strategy team and should only be able to look at lap and race data, never change any of it. Set up their access accordingly.

75. Two team principals accidentally try to register the same car number for two different drivers this season. What should stop this from happening?